

AMATH 582: Home Work 1

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This report focuses on tracking the movement of a submarine through the acoustic data it emits. In this paper, we will try to convert acoustic data into frequency domain by FFT method, and find dominant frequency by analyzing frequency domain data. Then, based on this dominant frequency, an appropriate Gaussian Filter is designed to reduce the influence of noise, extract the main running trajectory of submarine, and track it.

Introduction and Overview

Sonar localization is a common and effective method in military field, this paper will try to restore this process.

The data set selected in this paper is the unknown acoustic frequency of a submarine that is moving in the Puget Sound. This data set is a complex number set of 262144×49 . Among them, 262144 represents a $64 \times 64 \times 64$ spatial domain to represent the location where the acoustic frequency data was obtained, and 49 represents the time, more specifically, the data set is a 24-hour continuous data with half an hour intervals. Each data in the data set expresses the amplitude and phase of the acoustic in the form of a complex number, which should obey the following formula

$$P(x, y, z, t) = A(x, y, z)e^{i\phi(x, y, z)}, [1]$$

In which, function $A(x, y, z)$ represents amplitude, function $\phi(x, y, z)$ represents the phase and i is the imaginary unit.

Although this data has given a lot of information for general observation of submarine dynamics, more noise interferes with further analysis. Therefore, this paper needs to de-noise it to track the submarine more accurately

Theoretical Background

In this report, the FFT (Fast Fourier Transformation) method is mainly used to convert the spatial acoustic signal into the frequency domain, so as to obtain some meaningful information, such as the dominant frequency. The implementation of FFT is based on the following formulas^[2]:

$$FFT: F(k_x, k_y, k_z) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} \sum_{z=0}^{N_z-1} f(x, y, z) \cdot e^{-i2\pi(\frac{k_x x}{N_x} + \frac{k_y y}{N_y} + \frac{k_z z}{N_z})}$$

$$IFFT: f(k_x, k_y, k_z) = \frac{1}{N_x N_y N_z} \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} \sum_{z=0}^{N_z-1} F(x, y, z) \cdot e^{i2\pi(\frac{k_x x}{N_x} + \frac{k_y y}{N_y} + \frac{k_z z}{N_z})}$$

In which, k_x, k_y, k_z grids the frequency domains. And $k_{x,y,z} = \frac{L_{x,y,z}}{2\pi} \cdot (-\frac{N_{x,y,z}}{2}, \dots, -1, 0, 1, 2, \dots, \frac{N_{x,y,z}}{2} - 1)$. Therefore, distinct features of the signal can be isolated and the dominant frequency can be found by tracking on the maximum point. (i.e. the point (k_x^*, k_y^*, k_z^*) which makes $F(k_x, k_y, k_z)$ the largest).

After that, a Gaussian Filter will be designed to extract this center frequency in order to denoise the data and determine a more robust path of the submarine. It is defined by the following formula:

$$G(k_x, k_y, k_z) = e^{-\frac{1}{2\sigma^2}((k_x - k_x^*)^2 + (k_y - k_y^*)^2 + (k_z - k_z^*)^2)}$$

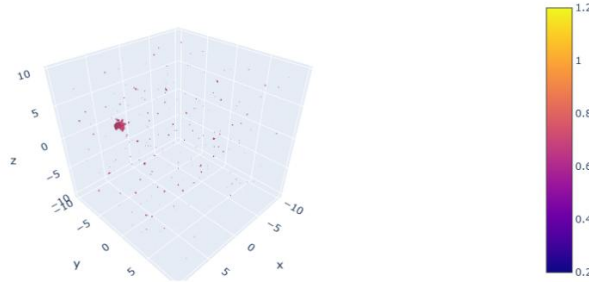
In which, (k_x^*, k_y^*, k_z^*) is the center point we got, and σ^2 denotes the variance, which controls the width of the filter. This filter can be multiplied with the original function, which will magnify the weights of the central point, the dominant frequency, and reduce the weights from the center outward, making the function smoother. In addition, if most concentration is put on the central point, which indicates that a large variant is chosen and makes the filter narrow, some low frequency information might be lost. That leads to a good result to get This is a good result to help us remove noise interference and locate the dominant frequency.

Algorithm Implementation and Development

In this report, numpy package has been used to process data, including taking the absolute value, mean value and maximum value. This completes the preparation and analysis of FFT. Also, the fftn and ifftn packages in numpy are also used to implement FFT method to the origin data and get back to the spatial domain from the frequency domain after using the filter. What's more, for the visualization, the plotly package is also included.

Computational Results

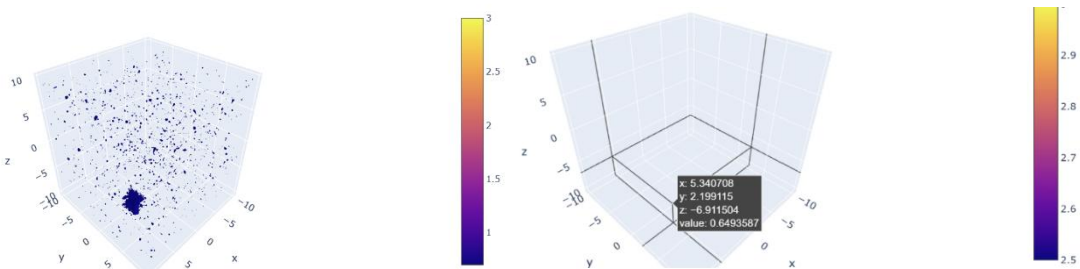
Before starting the computation, it's worth looking at the raw data we already have to obtain some information. It's a data based on the spatial domain, where $x, y, z \in [-10, 10)$ with an equal interval $\frac{20}{64}$. If we reshape the data into a $49*64*64*64$ set and cut them into slices as a function of time, we can get some static plots based on the absolute value of the data. For instance, here's a graph of the acoustic pressure measurements in 3D at the start time $t = 0$:



The figure of absolute value of acoustics pressure data at $t = 0$

Such figure shows all the points which are larger than 0.7, the figures at other time are similar. Through the figure we can obtain that although there is a roughly clustered high-amplitude point, a lot of noise fills the space, making it difficult to locate the central point we want.

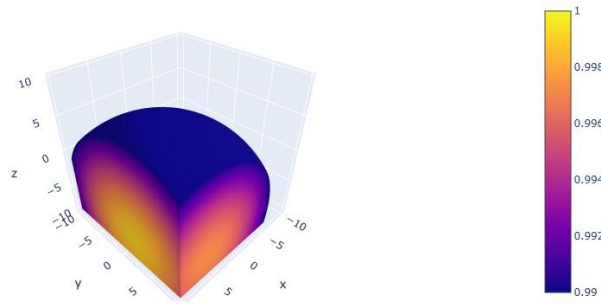
Therefore, we need to do the Fast Fourier Transformation to transform the data into the frequency domain and find the dominant frequency. Again, we cut the data into slices as a function of time, and do the 3D-Fast Fourier Transformation. As a result, we can still obtain a $49*64*64*64$ data set on the frequency domain, where the domain is $k_{x,y,z} = \frac{20}{2\pi} \cdot (-32, \dots, -1, 0, 1, 2, \dots, 31)$. Then we take an average in the time dimension to get a $64*64*64$ data. That means that we take the average of the 49 time slices for each elements in the matrix of the time slices. As is mentioned in the theoretical background, the dominant frequency we are looking for is the point whose absolute value is the maximum. Such maximum can be easily shown in the following figure, the point (5.341, 2.199, -6.911) is the only point that is larger than 2.5, which indicates that it is the maximum point.



The figure of the averaging in frequency domain and the maximum point in the averaged frequency domain

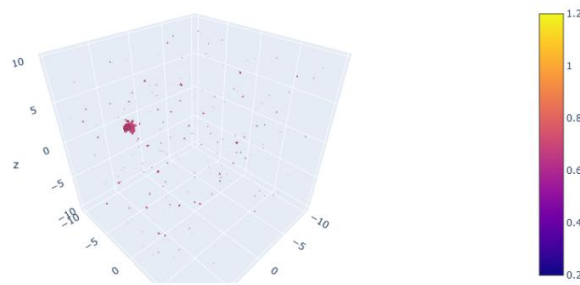
The first figure showed the averaging of the frequency domain, we showed all the points which is larger than 0.7 and smaller than 3, it has a lot of noise, but a dominant frequency is clear. And the second figure we found a maximum point in the averaged frequency domain, through this maximum point, we can obtain the dominant frequency, and track the frequency. Therefore, taking the point as the central point, the Gaussian Filter can be defined as $G(k_x, k_y, k_z) =$

$e^{-\frac{1}{2\sigma^2}((k_x-5.341)^2+(k_y-2.199)^2+(k_z+6.911)^2)}$. Taking the Gaussian Filter with variance $\sigma = 100$ as an example, here's its figure:



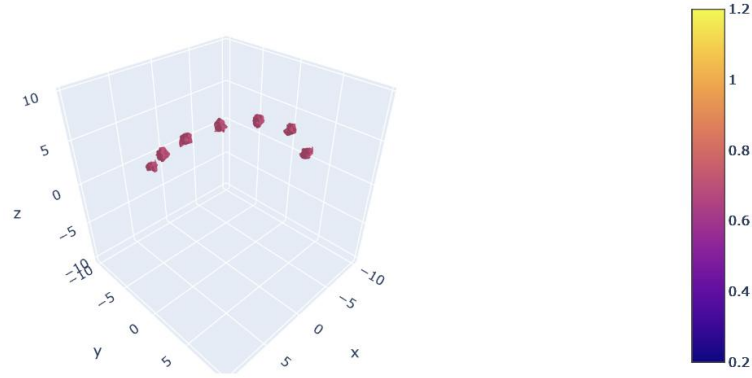
The figure of Gaussian Filter ($\sigma = 100$)

It's clear that the Gaussian Filter has the greatest value at the center point and reduces from the center outward, which stays the same as the theory. If we multiply it with the data in frequency domain and bring it back to the spatial domain, we are able to get a figure like this at the start time($t=0$):



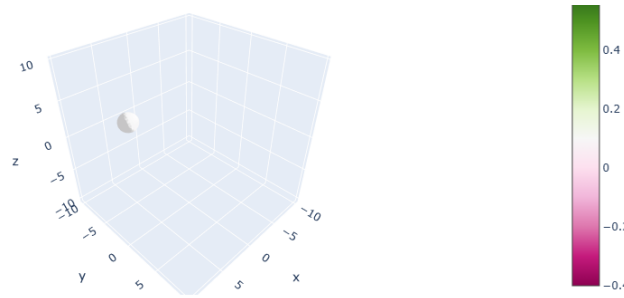
The figure of 'denoised' signal at the start time ($t=0$) with large variance ($\sigma = 100$)

It doesn't seem to be effective at noise reduction. We hypothesize that this is because the σ is so large that the gaussian filter does not weight the high frequency signal and the signals that are far from the center point much differently. Therefore, we are going to try on a smaller variance $\sigma = 10$. Here are some results of the denoised signals:



The figure of dynamic 'denoised' signal with variance $\sigma = 10$

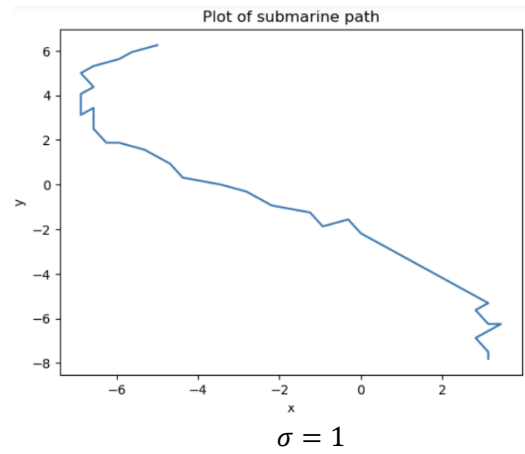
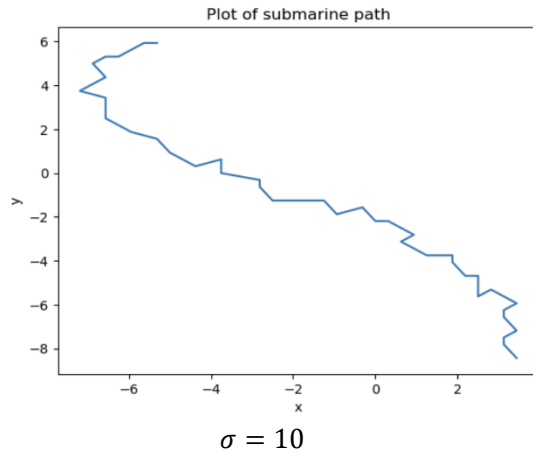
The figure shows the trace of the submarine at every 3.5 hour, i.e. at $t=0$, $t=7$, $t=14$, $t=21$, $t=28$, $t=35$, $t=42$ and $t=49$. Through the figure, we can clearly view that the noise are reduced and a more robust path of the submarine is able to be obtained. Thus, such variance is a proper one. Further, if the variance σ keeps reducing and becomes to 1, the result will become to:



The figure of 'denoised' signal at $t=0$ when variance $\sigma = 1$

This figure indicates that the path of the submarine becomes even more robust, however, some information is missing, since we only pay close attention to the highest frequency and weigh it too much. Although all the data away from the highest frequency are ignored in such situation, it seems that it doesn't influence our tracking to the center frequency, which means that the result can also be accepted.

Finally, we can use the result of the denoised signal data with $\sigma = 10$ and 1 to locate the submarine in x-y plane. Here's the result:



The plot of 'denoised' submarine path at $t=0$ with $\sigma = 10$ and 1

The track is similar and errors are acceptable, although when $\sigma = 1$ the track seems to be smoother.

Summary and Conclusions

To draw a conclusion, in this report we succeed in finding the highest frequency, which might be the frequency caused by the submarine by using FFT method. Then a Gaussian Filter has been designed to track this frequency, and denoise the signal. After using the Gaussian Filter, we transfer the frequency back to the spatial domain and effectively find the path of the submarine.

Acknowledgements

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References

- [1] Hung J W, Hsieh H J, Chen B. Robust speech recognition via enhancing the complex-valued acoustic spectrum in modulation domain[J], 2015, 24(2): 236-251.
- [2] S. Holm, "Optimum FFT-based frequency acquisition with application to COSPAS-SARSAT", April 1993.